

Trilaterales Transferprojekt DFG – Fraunhofer

APEM

February 8, 2024

Agenda

10:00 – 10:40: Pricing Rules & Power Flow Relaxations

10:40 – 11:00: Framework Design

11:00 - 12:00: Bidding Zone Review, Local Incentives, TenneT Tool Chains

12:00 - 13:30: Lunch Break

13:30 - 14:30: European Electricity Markets, EUPHEMIA Algorithm

14:30 - 16:00: Discussion

- 14:30 – 14:45: Alternative Price Zone Configurations for Germany
- 14:45 – 15:00: Integrating Renewables into Nodal Energy Markets
- ...

Electricity Markets – Pricing Rules and Power Flow Models

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Clearing and Pricing in Electricity Markets is Hard

Two main sources of complexity:



Non-Convex Cost Functions

- Mixed-integer allocation problem $\rightarrow$ NP-hardness
- Non-existence of Walrasian equilibria
- Violation of efficiency, individual rationality, or budget balance



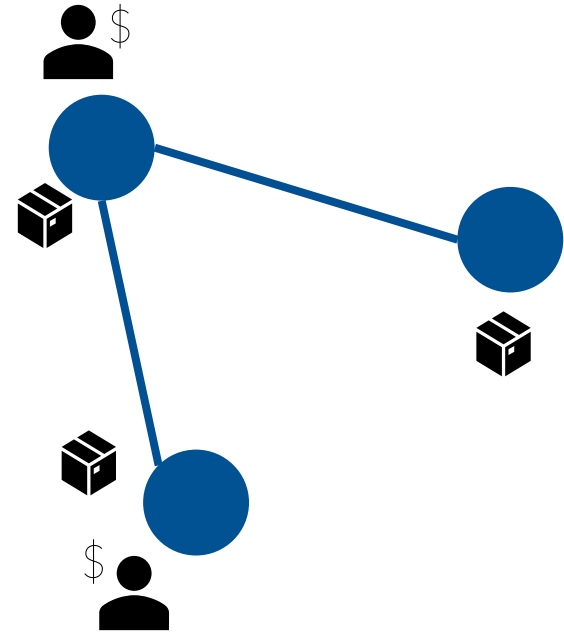
Transmission Constraints

- Physical laws of power flow $\rightarrow$ non-linear and non-convex constraints
- Computationally intractable allocation problem
- Ex-post adjustment of solution to relaxed allocation problems

Setup and Notation

Our market consists of:

- 1) A set of nodes N .
(and a set of power lines L encoded as pairs of nodes (i, k))
- 2) A set of items corresponding to real or reactive power at each node
(encoded as complex number $x_i = \text{Re}(x_i) + j \text{Im}(x_i)$)
- 3) A set of buyers B .
(with valuations $v_b: \mathbb{C}^{|N|} \rightarrow \mathbb{R}$)
- 4) A set of sellers S .
(with costs $c_s: \mathbb{C}^{|N|} \rightarrow \mathbb{R}$)



Welfare Maximization Problem

Define the following welfare maximization problem:



Non-Convex Cost Functions

$$\max_{x,y} \sum_b v_b(x_b) - \sum_s c_s(y_s)$$

s. t. $x, y \in \Psi$



Transmission Constraints

Competitive Equilibrium Theory

- For a convex market, the *Welfare Theorems* (Arrow & Debreu 1954) hold.
- A primal solution x^* and a dual solution p^* assemble to form a *Walrasian equilibrium*.

Such a pair (x^*, p^*) satisfy certain desirable properties:



Efficiency

→ Primal Optimality



Supply-Demand
Balance

→ Primal Feasibility



Envy-Freeness

→ Strong duality

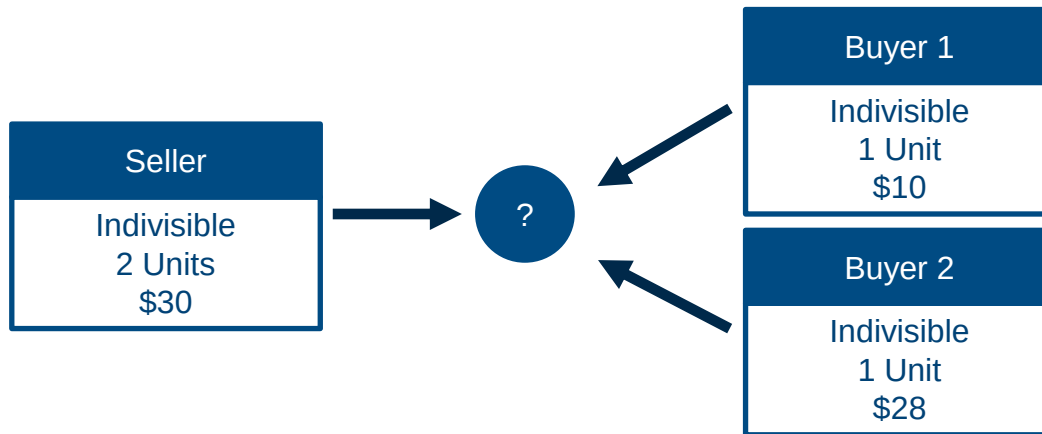


Budget Balance

→ C. Slackness

Competitive Equilibrium Theory

With *non-convexities* present, a Walrasian equilibrium might not exist.



Clearing the market



- Seller makes a loss at $p \leq \$10$
- Buyer 1 makes a loss at $p > \$10$

Electricity market pricing in practice

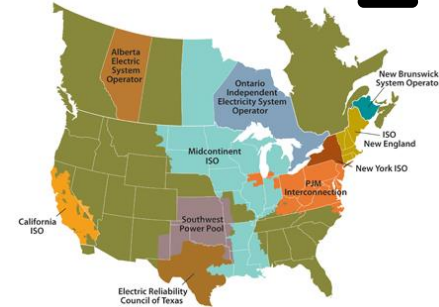
In the US:

Efficient outcome, linear prices + uplifts / penalties.

In the EU:

Suboptimal outcome, linear prices, no side payments.

Primal-dual problems coupled, becoming intractable.



Competitive Equilibrium Theory

Different pricing rules exist (Liberopoulos & Andrianesis, 2016) and are often benchmarked by several measures of deviation from Walrasian equilibria.

- **Lost opportunity costs (LOCs):** Forgone profits of market participants (*global* or *local*)

$$LOC_s = \hat{u}_s(\lambda) - u_s(y_s^*|\lambda) = (\max_{y_s} \lambda^T \text{Re}(y_s) - c_s(y_s)) - (\lambda^T \text{Re}(y_s^*) - c_s(y_s^*))$$

- **Make-whole payments (MWPs):** Compensations for negative profits of market participants

$$MWP_s = \max(-u_s(y_s^*|\lambda), 0) = \max(-(\lambda^T \text{Re}(y_s^*) - c_s(y_s^*)), 0)$$

- **Potential congestion revenue shortfall (PCRS):** LOCs for transmission operators

$$PCRS = \max_{x,y \in \Psi} \lambda^T (\text{Re}(x_N) - \text{Re}(y_N)) - \lambda^T (\text{Re}(x_N^*) - \text{Re}(y_N^*))$$

Competitive Equilibrium Theory

We focus on two pricing rules:

Integer Programming (IP) Pricing (O'Neill et al., 2005)

Idea:

- 1) Solve welfare maximization problem to optimality
- 2) Fix integer variables to their optimal values
- 3) Retrieve prices from duals of resulting LP

Properties:

- „Marginal pricing“
- Minimizes PCRS under DC but high MWPs

Convex Hull (CH) Pricing (Hogan & Ring, 2003)

Idea:

- 1) Replace non-convex cost functions by convex envelopes
- 2) Retrieve prices from duals of resulting LP

Properties:

- Minimizes LOCs
- Computationally hard in general

Join Pricing Rule

Dual pricing problem: $\min_p \sum_i \max\{MWP_i(p), LLOC_i(p)\}$

Pros:

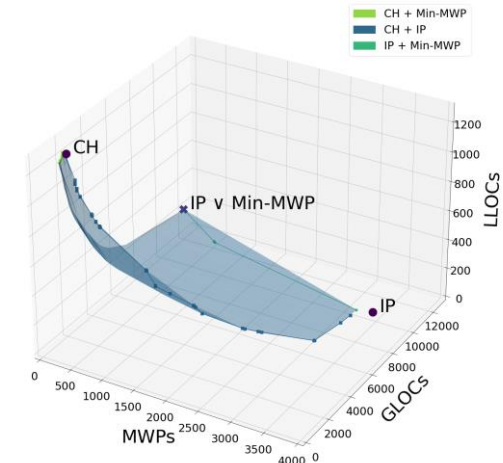
- + Favorable balance between MWP and PCRS.
- + Poly-time (LP).
- + Accurate incentives.

Cons:

- Not Pareto optimal. (but less MWPs than IP)

	GLOCs [\$]	MWPs [\$]	LLOCs [\$]
CH	1436.21	202.57	1272.12
IP	10364.24	3387.00	0
Min-MWP	2.1×10^{11}	0	1.1×10^{11}
0.9 IP + 0.1 Min-MWP	11025.03	693.00	111.79
0.9 CH + 0.1 Min-MWP	1432.93	199.27	1119.85
0.5 IP + 0.5 CH	1570.13	606.21	729.29
IP v Min-MWP	12022.38	0	191.14

Table 4: RTS System - GLOCs, MWPs, LLOCs



Transmission Constraints

Power flow models describe the relationship between power injections / withdrawals at each node and the resulting line flows (Cain & O'Neill, 2012; Molzahn & Hiskens, 2019).

Core problem: **Alternating Current Optimal Power Flow (ACOPF)** problem

$$\Psi^{AC} = \{x \in \mathbb{C}^{|N| \times |B|}, y \in \mathbb{C}^{|N| \times |S|} :$$

$$\sum_{s \in S} \text{Re}(y_{is}) - \sum_{b \in B} \text{Re}(x_{ib}) = \sum_{k \in N} V_{di} (G_{ik} V_{dk} - B_{ik} V_{qk}) + V_{qi} (B_{ik} V_{dk} + G_{ik} V_{qk}) \quad \forall i \in N,$$

$$\sum_{s \in S} \text{Im}(y_{is}) - \sum_{b \in B} \text{Im}(x_{ib}) = \sum_{k \in N} V_{di} (-B_{ik} V_{dk} - G_{ik} V_{qk}) + V_{qi} (G_{ik} V_{dk} - B_{ik} V_{qk}) \quad \forall i \in N,$$

$$V_{di}^2 + V_{qi}^2 = |V_i|^2 \quad \forall i \in N,$$

$$(V_i^{\min})^2 \leq |V_i|^2 \leq (V_i^{\max})^2 \quad \forall i \in N,$$

$$V_i \in \mathbb{C} \quad \forall i \in N \quad \}.$$

Non-linear and non-convex constraints

→ Intractable for practical problem sizes

Convex Relaxations for ACOPF

- **ACOPF**
 - Accurate modeling of the network physics
 - Computationally intractable



Convex Relaxations for ACOPF

- ACOPF
- **Semidefinite Programming (SDP) Relaxation** (Shor, 1987; Lavaei & Low, 2012)
 - Equivalent formulation for ACOPF as rank-constrained SDP
 - Removing rank constraint yields convex SDP relaxation
 - Convexity but computational hardness

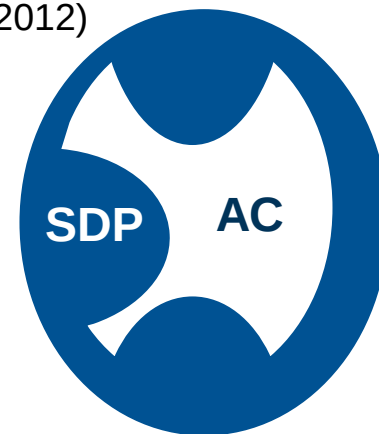
$$\Psi^{SDP} = \{x \in \mathbb{C}^{|N| \times |B|}, y \in \mathbb{C}^{|N| \times |S|} :$$

$$\sum_{s \in S} \text{Re}(y_{is}) - \sum_{b \in B} \text{Re}(x_{ib}) = \langle L_{P_i}, W \rangle \quad \forall i \in N,$$

$$\sum_{s \in S} \text{Im}(y_{is}) - \sum_{b \in B} \text{Im}(x_{ib}) = \langle L_{Q_i}, W \rangle \quad \forall i \in N,$$

$$|V_i|^2 = \langle M_i, W \rangle \quad \forall i \in N,$$

$$W \succcurlyeq 0 \}.$$



Convex Relaxations for ACOPF

- ACOPF
- SDP Relaxation
- **Second-Order Conic (SOC) Relaxation** (Jabr, 2006; Gan et al., 2012)
 - Reformulate and condense non-convexities in single equality constraint
 - Replacing equality with inequality yields convex SOC relaxation
 - Efficient interior-point method solvers, also in mixed-integer versions

$$\Psi^{SOC} = \{x \in \mathbb{C}^{N \times |B|}, y \in \mathbb{C}^{N \times |S|} :$$

$$\sum_{s \in S} \text{Re}(y_{is}) - \sum_{b \in B} \text{Re}(x_{ib}) = G_{ii}c_i + \sum_{k \in N \setminus \{i\}} G_{ik}cr_{ik} - B_{ik}ci_{ik} \quad \forall i \in N,$$

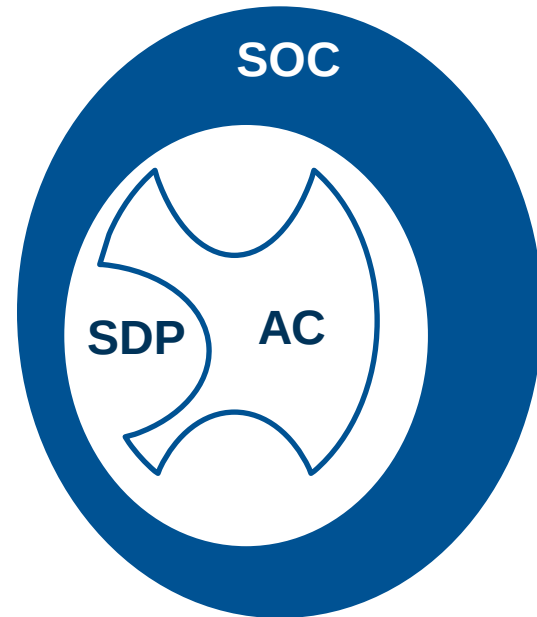
$$\sum_{s \in S} \text{Im}(y_{is}) - \sum_{b \in B} \text{Im}(x_{ib}) = -B_{ii}c_i + \sum_{k \in N \setminus \{i\}} -B_{ik}cr_{ik} - G_{ik}ci_{ik} \quad \forall i \in N,$$

$$cr_{ik} = cr_{ki} \quad \forall (i, k) \in L,$$

$$ci_{ik} = -ci_{ki} \quad \forall (i, k) \in L,$$

$$cr_{ik}^2 + ci_{ik}^2 \leq c_i c_k \quad \forall (i, k) \in L,$$

$$c_i, cr_{ik}, ci_{ik} \in \mathbb{R} \quad \forall (i, k) \in L \quad \}.$$



Convex Relaxations for ACOPF

- ACOPF
- SDP Relaxation
- SOC Relaxation
- **Quadratic Convex (QC) Relaxation** (Hijazi et al., 2017)
 - Derived from polar representation of ACOPF
 - Based on convex envelopes for square, bilinear, sine, cosine terms

$$\Psi^{QC} = \Psi^{SOC} \cap \{x \in \mathbb{C}^{N \times |B|}, y \in \mathbb{C}^{N \times |S|} :$$

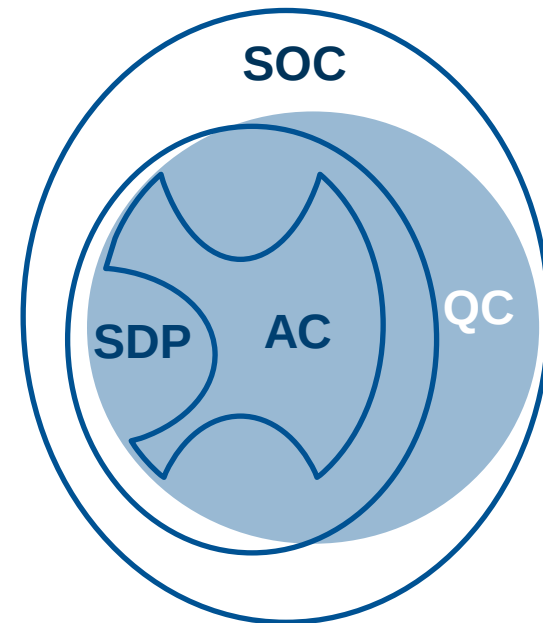
$$c_i \in \langle |V_i|^2 \rangle \quad \forall i \in N,$$

$\langle \cdot \rangle$ denotes
convex envelope

$$cr_{ik} \in \langle \langle |V_i| |V_k| \rangle \langle \cos(\theta_i - \theta_k) \rangle \rangle \quad \forall (i, k) \in L,$$

$$ci_{ik} \in \langle \langle |V_i| |V_k| \rangle \langle \sin(\theta_i - \theta_k) \rangle \rangle \quad \forall (i, k) \in L,$$

$$c_i, cr_{ik}, ci_{ik}, \theta_i \in \mathbb{R} \quad \forall (i, k) \in L \quad \}.$$



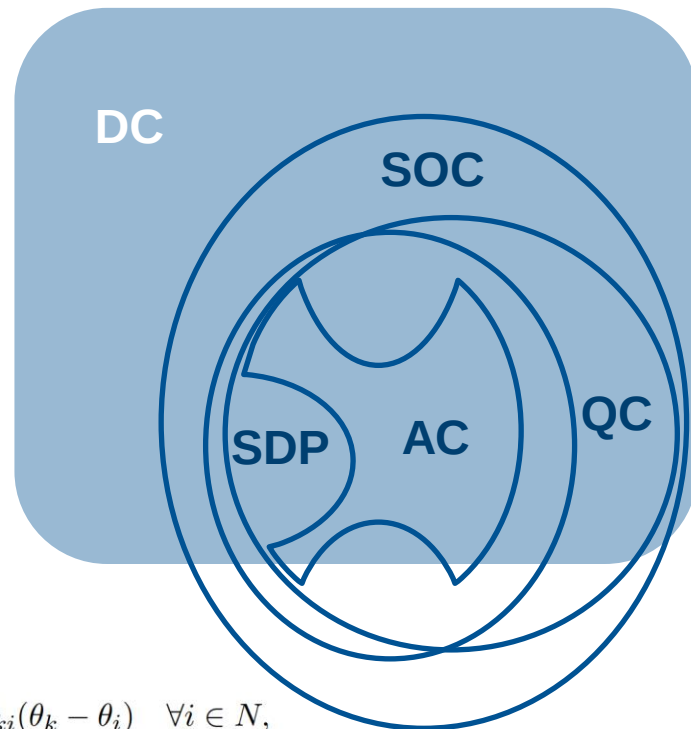
Convex Relaxations for ACOPF

- ACOPF
- SDP Relaxation
- SOC Relaxation
- QC Relaxation
- **Direct Current (DC) Approximation** (Stott et al., 2009)
 - Fully linearized version
 - Line resistances and reactive power are ignored

$$\Psi^{DC} = \{x \in \mathbb{C}^{|N| \times |B|}, y \in \mathbb{C}^{|N| \times |S|} :$$

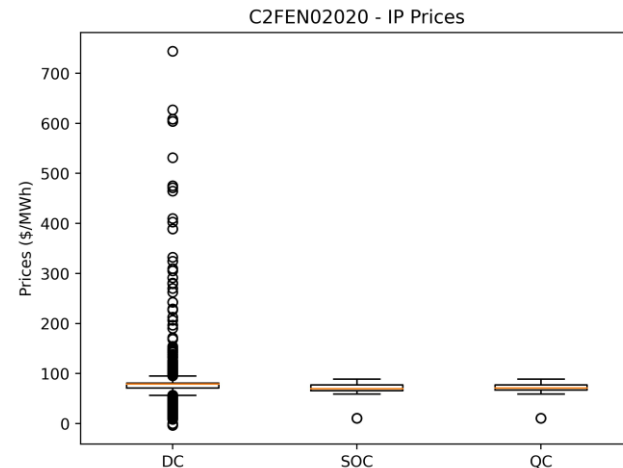
$$\sum_{s \in S} \text{Re}(y_{is}) - \sum_{b \in B} \text{Re}(x_{ib}) = \sum_{k \in N: (i,k) \in L} -B_{ik}(\theta_i - \theta_k) - \sum_{k \in N: (k,i) \in L} -B_{ki}(\theta_k - \theta_i) \quad \forall i \in N,$$

$$\theta_i \in \mathbb{R} \quad \forall i \in N \quad \}.$$



First Experimental Results - ARPA-E Grid Challenge II

- The DC model – as it is currently used – can lead to high and unjustified price peaks
- Allocations vary significantly between different power flow models, affecting pricing and required redispatch
- Tighter power flow relaxations imply smaller deviations from Walrasian equilibria for the final AC-feasible allocation
- The computational costs remain as limiting factor for tighter power flow relaxations



Conclusions

Clearing electricity markets is a complex problem:

- 1) Non-convex valuation or cost functions → Non-existence of Walrasian equilibria
- 2) Non-linear non-convex transmission constraints → Computational intractability

APEM Proposal – Scientific Goals:

In particular, we want to address two main questions in this transfer project:

- *Which optimization models best approximate the underlying AC optimal power-flow problem and allow for the computation of realistic problem sizes?*
- *Which pricing rules best achieve design desiderata, such as low side-payments, adequate congestion signals, and low lost opportunity costs?*

Thanks!